Nizar Esmat Fawzy

Software Engineer
Cairo, Egypt | +20 109 258 9504 | nizaresmat2000@gmail.com
linkedin.com/in/nizar-esmat | github.com/Nizar-Esmat | Portfolio

PROFESSIONAL SUMMARY

Backend-focused Software Engineer with 2 years of experience designing and shipping production REST APIs. Builds Python/Django REST Framework and NestJS services on PostgreSQL, owning API contracts, schema design, query optimization, and automated tests across 50+ endpoints and 5 production platforms in a 10+ engineer team. Works daily in Linux, Docker, and Git.

EDUCATION

Nahda University, Beni-Suef, Egypt **Sep 2019 - Jul 2023**
Bachelor of Computer Science

Information Technology Institute (ITI), Smart Village, Egypt **Nov 2024 - Mar 2025**
Full Stack Development Track (Python)

- Completed a competitive full-stack Python program: built RESTful APIs with Django and **Django REST Framework** over a normalized PostgreSQL schema designed for 10k+ users, plus ETL-style data workflows.

PROFESSIONAL EXPERIENCE

Opream, Cairo, Egypt **May 2025 - Present**
Software Engineer

- Build and maintain Python/**Django REST Framework** services on PostgreSQL: model and serializer design, viewsets with role-based permission classes, token authentication, and Django ORM query tuning behind versioned REST endpoints.
- Design and ship 50+ RESTful API endpoints across 5 production platforms, owning each feature from API contract and relational data model through testing and release; stack spans Django, NestJS/TypeScript, PostgreSQL, and React.
- Architected **Mostaqbel Watan**, a private social platform serving 1,000+ active users, with Socket.io messaging, Jitsi meetings, and an automated Facebook Graph API ingestion pipeline; also delivered ride-sharing (Fawry payments), lesson management, and event ticketing platforms.
- Cut query latency by normalizing schemas, writing and optimizing complex SQL, adding targeted indexes, and eliminating N+1 patterns; cover critical paths with pytest and Jest unit and integration tests, develop against Dockerized services, and review teammates' pull requests.

iSchool, Cairo, Egypt **Jul 2024 - Nov 2024**
Software Instructor (Part-time)

- Taught programming fundamentals and algorithmic problem solving, breaking down core CS concepts for students.

WE School, Cairo, Egypt **Aug 2023 - Jun 2024**
Web Development Instructor

- Taught web development with HTML, CSS, and JavaScript, guiding students from fundamentals to working projects.

PROJECTS

ECPC Movement Control | *NestJS, PostgreSQL, TypeORM, Socket.io, React, JWT* GitHub

- Built a real-time volunteer dispatch system for the Egyptian Collegiate Programming Contest: a scoring algorithm auto-assigns requests by hall proximity, gender match, and idle time, backed by a versioned PostgreSQL schema, an audit trail, and live updates under JWT-secured roles.

Chat Application | *React, Node.js, Express.js, Socket.io, MongoDB, Cloudinary* GitHub Postman

- Hardened a full-stack real-time chat API with HTTP-only cookie JWT authentication, Arcjet rate limiting, and Cloudinary media storage over HTTPS; Socket.io WebSocket transport carries private messaging, group channels, and live presence to a React/Zustand client.

Smart Learning Management System (Graduation Project) | *Python, scikit-learn, pandas, NumPy*

- Owned the AI module of a team-built learning management system: a linear regression model, trained on cleaned historical academic records, that predicts a student's final grade from midterm and coursework performance so instructors can flag at-risk students early.

TECHNICAL SKILLS

Programming Languages: Python, SQL, TypeScript, JavaScript, C++
Backend: Django, Django REST Framework, REST API Design, NestJS, Node.js, Express.js, Socket.io
Databases: PostgreSQL, MongoDB, Complex SQL, Data Modeling, Indexing, Query Optimization
DevOps & Testing: Docker, Linux/UNIX, Nginx, Gunicorn, Git, GitHub, pytest, Jest, Unit & Integration Testing
Networking & Security: TCP/IP, HTTP/HTTPS, TLS/SSL, WebSockets, JWT, Authentication & Authorization
Computer Science: Data Structures & Algorithms, OOP, SOLID, Design Patterns, System Design
Data & ML: scikit-learn, pandas, NumPy, Regression Modeling, Data Cleaning
Frontend: React, HTML, CSS **Spoken Languages:** Arabic (Native), English (Fluent)

CERTIFICATIONS

Python Professional Track – ITI | SQL & PostgreSQL – Udemy | Node.js – Udemy | MERN Stack – Route Academy

VOLUNTEERING & ACHIEVEMENTS

2nd Place – Nahda University ECPC Qualifiers (2021, 2022); **Organizing Volunteer** – ICPC/ECPC/ACPC, 2022-2024.

300+ algorithmic problems solved in C++ on Codeforces – data structures, algorithms, and complexity analysis.